

Weak Error Analysis and Small-Noise Optimization of Stochastic Linear Multistep Methods (SLMMs) for Itô and Lévy SDEs

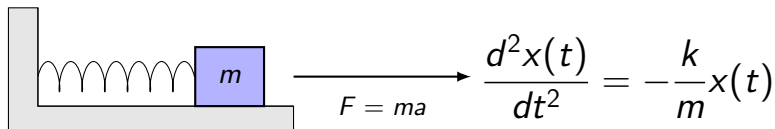
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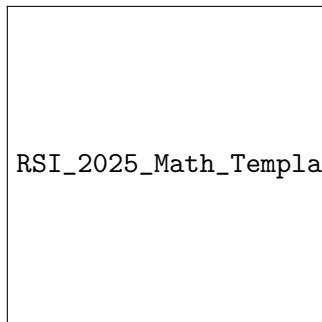
Deterministic Processes

We know how to model deterministic processes.



Brownian Motion (Wiener Process)

Real-world processes have randomness.



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Figure 1: Particle Interaction System

Ordinary Differential Equation (ODE):

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Stochastic Correction:

$$dX_t = f(X_t, t)dt + g(X_t, t)dW_t.$$

We use X_t instead of $x(t)$ to denote a *stochastic process*.

Example (Black-Scholes Equation)

For a stock price X_t ,

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

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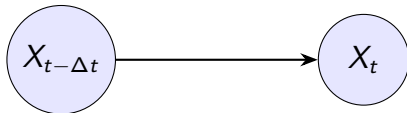
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The Problem

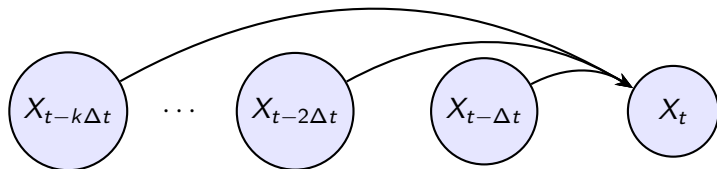
Goal: Approximate solution X_t with process $\bar{X}_t$ “close” in distribution at a low computational cost.

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One-Step Methods



Multi-Step Methods



One-Step Euler-Maruyama:

$$\bar{X}_{t+\Delta t} = \bar{X}_t + f(\bar{X}_t, t)\Delta t + g(\bar{X}_t, t)\Delta W_t.$$

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Two-Step Generalization:

$$\bar{X}_{t+\Delta t} = \bar{X}_t + \Delta t \left(\frac{3}{2}f(\bar{X}_t, t) - \frac{1}{2}f(\bar{X}_{t-\Delta t}, t - \Delta t) \right) + g(\bar{X}_t, t)\Delta W_t.$$

A "Closeness" Metric

Weak error expansion:

$$\mathbb{E}[\varphi(X_T)] - \mathbb{E}[\varphi(\bar{X}_T)] \sim C_{\varphi,\rho} \Delta t^{\rho} + C_{\varphi,\rho+1} \Delta t^{\rho+1} + \dots$$

with smooth φ .

- Constant Factor Improvement: Decrease $C_{\varphi,\rho}$
- Asymptotic Improvement: Increase ρ

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- **Should we use multistep methods in practice?**

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There is a surefire method to compute an arbitrarily high accuracy weak error expansion of any linear multistep method.

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⇒ **Constant Factor Improvement**

Key Idea: Correspondence between Weak Error of SDEs and Elliptic PDEs given by Feynman-Kac Theorem.

Results

Theorem (A., Lee-2025)

When the noise is “small” one can construct multistep methods that make **asymptotic improvement** in error over their one-step analogues.

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Results

We have analogous results for Lévy-driven SDEs.

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Results

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We should use multistep methods in practice.

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